

Improving Continual Learning Robustness in Medical Imaging via Illumination Adaptive Transformer

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Abstract

Continual learning (CL) refers to the capability of a model to learn progressively from an evolving stream of data, retaining previously acquired knowledge while integrating new information. This capability is pivotal in advancing medical image classification, especially when data availability fluctuates. Beyond investigating CL performance under standard clean-data conditions, this paper systematically evaluates the robustness of representative CL strategies in *uncertain imaging contexts*, where visual quality is degraded by varying degrees of low-light conditions and over- or under exposure. In this paper, we augment the training and evaluation data with controlled, simulated low-light and contrast perturbations to model these uncertain conditions, which mimic real-world variability frequently encountered in clinical acquisition environments. Our method integrates an automatic illumination calibration module, termed the Illumination Adaptive Transformer (IAT), within existing CL frameworks to mitigate the adverse effects of such degradations. This module dynamically adjusts the image illumination and contrast, aiming to enhance the visibility of critical features in a data-driven, end-to-end manner without requiring manual tuning or image-specific heuristics. Experiments demonstrate that incorporating the IAT module consistently improves final classification accuracy and robustness across multiple continual learning strategies under all simulated uncertainty levels on the PathMNIST dataset.

Keywords: continual learning; medical image classification; uncertain contexts; illumination adaptive transformer

1. Introduction

The field of medical imaging is experiencing rapid growth, accompanied by an increasing demand for accurate and versatile classification models. As the spectrum of pathologies requiring identification continues to expand, there is a critical need for systems that can adapt to newly emergent conditions and evolving class distributions without compromising their performance on previously learned tasks. This requirement directly motivates the adoption of continual learning (CL) paradigms in medical image analysis. Continual learning refers to a model's capability to incrementally learn from an evolving data stream while retaining previously acquired knowledge and integrating new information. However, conventional model adaptation strategies, such as fine-tuning on newly collected data batches, frequently lead to catastrophic forgetting, where updated representations overwrite previously learned decision boundaries, resulting in significant performance degradation on earlier tasks.

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Several practical constraints further impede the straightforward solution of retraining models from scratch using all available data. In healthcare environments, data privacy regulations and institutional policies often limit access to historical patient images, preventing centralized storage and comprehensive data reuse for retraining. Meanwhile, the exponential growth of modern medical imaging repositories substantially increases computational cost, energy consumption, and storage requirements, rendering repeated full-scale retraining economically and environmentally impractical. To address these challenges, recent studies in medical continual learning have explored rehearsal-based memory methods, regularization techniques, and parameter-isolation strategies to alleviate catastrophic forgetting under controlled clean-data conditions (7). While these approaches have demonstrated encouraging results, most evaluations assume stable acquisition environments and do not explicitly examine learning behavior under domain shifts caused by real-world imaging variability.

In parallel, a growing body of work on illumination restoration and enhancement has demonstrated that correcting visual degradations can significantly improve perceptual quality and downstream recognition performance across multiple medical modalities. Recent studies have investigated deep learning-based illumination correction for whole-slide histopathology images to address uneven background and shading artifacts (22; 23), global-local enhancement pipelines for low-light endoscopic images (24), and stripe artifact removal and illumination normalization in large-scale microscopy (26), as well as Bayesian devignetting approaches for retinal fundus imaging (27). Moreover, diffusion-based universal enhancement models have been proposed to correct illumination, contrast, and noise across diverse imaging modalities, demonstrating consistent benefits for downstream medical analysis (25). Lightweight transformer-based architectures such as the Illumination Adaptive Transformer (IAT) further offer efficient solutions for exposure correction in static datasets (20), while showing improvements in vision task robustness under adverse lighting conditions (13; 14; 15). However, these enhancement methods have largely been evaluated in offline or fixed-dataset settings and have not been systematically studied within the sequential learning paradigm of continual learning.

In this work, we propose a robust and adaptive medical image classification framework that integrates representative CL strategies with an automatic illumination calibration module. By explicitly evaluating continual learning performance under simulated low-light and exposure perturbations and investigating the role of illumination enhancement, our study provides empirical evidence on how enhancement modules can stabilize learning dynamics and improve robustness under uncertain acquisition conditions. Our findings advance the development of robust medical image analysis pipelines and offer actionable guidance for building more dependable and clinically deployable decision support systems.

2. Background and Related Work

2.1. Background on Continual learning

Continual Learning (CL) can be formally defined as follows (1). Let $\mathcal{D} = D_1, D_2, \dots, D_T$ denote a sequence of T learning tasks, where each task D_t is associated with a dataset $(x_i^t, y_i^t)_{i=1}^{N_t}$. Here, $x_i^t \in \mathcal{X}$ represents the input data sample and $y_i^t \in \mathcal{Y}$ denotes the corresponding class label for the i -th instance in task t .

A continual learning model parameterized by θ aims to learn a classification function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ that performs well not only on the current task but also on all previously encountered tasks. At each time step t , the model is trained using data from task D_t by solving the optimization problem:

$$\theta_t = \arg \min_{\theta} \mathcal{L}(f_\theta, D_t), \quad (1)$$

where $\mathcal{L}$ denotes an appropriate task-specific loss function.

The core challenge in CL is learning θ_t in a manner that balances plasticity and stability, ensuring that acquisition of new knowledge does not degrade previously learned representations. Formally, this requirement can be expressed as:

$$\mathbb{E}(x, y) \sim D_k [\mathcal{L}(f_{\theta_t}(x), y)] \leq \epsilon_k, \quad \forall k \leq t, \quad (2)$$

where ϵ_k is a small positive threshold enforcing sustained performance on all tasks observed up to time t . Beyond predictive performance, effective CL systems are expected to satisfy several practical criteria. Adaptability refers to the model’s ability to mitigate catastrophic forgetting and rapidly incorporate new knowledge. Generalizability denotes the capacity to accommodate both intra-task and inter-task data distribution shifts. Efficiency emphasizes the ability to update the learning model using minimal additional computational cost and limited memory overhead, making CL feasible under real-world constraints.

Existing CL strategies can be broadly categorized into five major families based on their primary mechanisms for handling catastrophic forgetting and task adaptation:

1. **Replay-based strategies:** These methods preserve information from past tasks by storing representative exemplars or generating pseudo-samples for rehearsal during training on new tasks. Representative examples include Experience Replay (ER) (21) and Generative Replay (8).
2. **Regularization-based strategies:** These approaches constrain parameter updates to prevent excessive drift from values deemed important for previous tasks. They include both:
 - Weight regularization methods, such as Elastic Weight Consolidation (EWC) (18), and
 - Functional regularization techniques, such as Learning without Forgetting (LwF) (7).
3. **Architectural strategies:** These methods dynamically modify network structures to accommodate new tasks while preserving existing knowledge. Examples include Progressive Neural Networks (16) and Dynamically Expanding Networks (10).
4. **Meta-learning strategies:** These techniques aim to learn model update rules that are intrinsically resilient to forgetting across tasks. Notable examples include Meta-Continual Learning (11) and Online-Aware Meta-learning (17).

5. **Hybrid strategies:** These approaches combine multiple paradigms to exploit complementary strengths. For instance, iCaRL (4) integrates replay mechanisms with regularization techniques.

In this work, we focus on three representative strategies covering both replay-based and regularization-based paradigms including Experience Replay (ER), Elastic Weight Consolidation (EWC), Synaptic Intelligence (SI). Specifically, each of strategy will be integrated with our proposed illumination calibration framework to evaluate their robustness under uncertain conditions.

2.2. Illumination Calibration in Medical Imaging

The varying light conditions in real-world acquisition environments present significant challenges for both human visual perception and downstream vision tasks (e.g., semantic segmentation and object detection). Images captured under poor lighting conditions suffer from limited photon counts and increased sensor noise, leading to degraded contrast and loss of fine structures critical for clinical interpretation. Conversely, scenes exposed to intense illumination, such as direct sunlight or strong artificial lighting, frequently experience saturation due to the limited dynamic range of image sensors and non-linear processing effects within the camera image signal processing pipeline.

Several techniques have been developed to address challenging lighting conditions, including low-light enhancement (3; 5; 6) and exposure correction(9; 12). Low-light enhancement methods aim to restore structural details while suppressing amplification of noise. Exposure correction techniques focus on adjusting underexposed or overexposed images to recover visually plausible outputs under extreme exposure durations. While the majority of prior work primarily targets perceptual image quality, recent studies have demonstrated that illumination-aware enhancement can also improve the robustness of high-level vision tasks, such as under low-light settings(13; 14; 15) and overexposure(19). In medical imaging specifically, deep learning-based illumination correction has been shown to improve the reliability of downstream analysis by compensating for non-uniform background and shading artifacts in whole-slide histopathology images(22; 23), enhancing low-light endoscopic images through joint global-local modeling(24), and learning modality-agnostic enhancement priors with diffusion-based models for diverse medical modalities(25). Furthermore, illumination calibration has been explored in microscopy and retinal imaging, where dedicated methods reduce stripe artifacts and uneven illumination in large-scale microscopy(26) and correct vignetting in fundus images via Bayesian estimation of the illumination field(27), thereby stabilizing subsequent segmentation and classification pipelines.

In 2022, Cui *et al.*(20) introduced a lightweight transformer-based image enhancement and exposure correction model, referred to as the Illumination Adaptive Transformer (IAT). Distinct from conventional image enhancement networks, IAT is explicitly designed for computational efficiency, comprising only approximately 90,000 parameters, which enables effective deployment on resource-constrained platforms such as laptops or smartphones. After the camera captures raw-RGB data, images are rendered into standard sRGB format through an image signal processor (ISP). Many existing illumination-adaptive approaches operate directly on raw-RGB data, limiting their direct applicability to widely available sRGB images. To address this limitation, IAT explicitly models the ISP pipeline

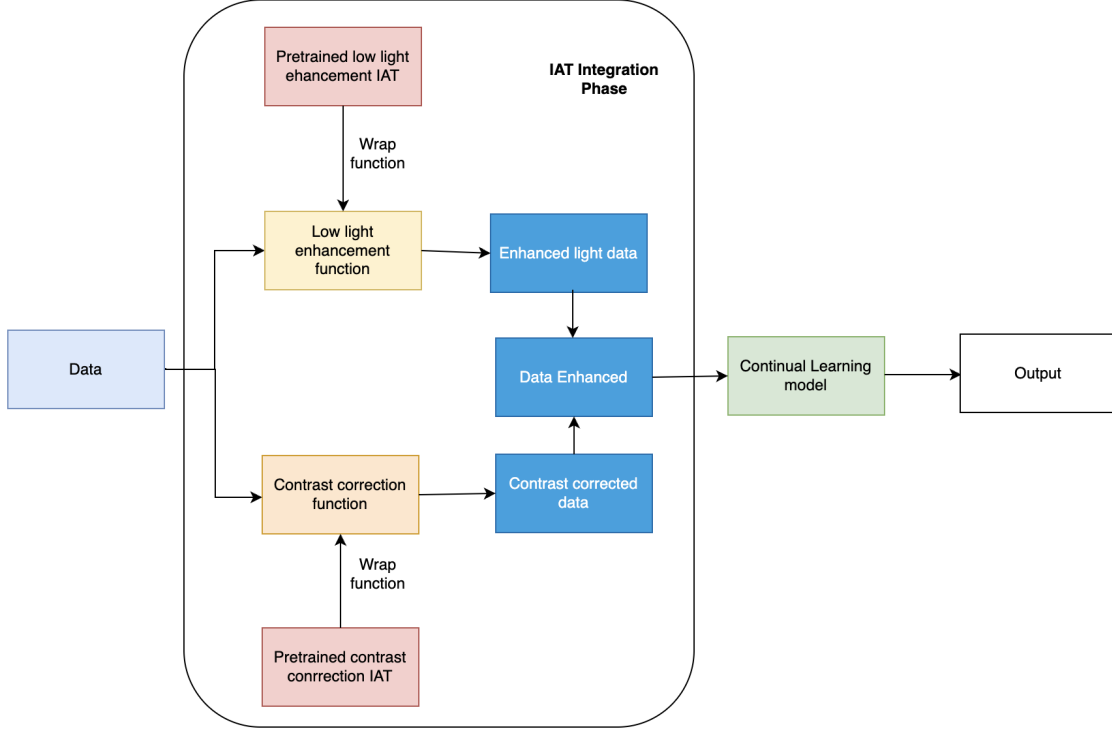


Figure 1: Overall workflow of our method. Incoming images are analyzed for illumination degradations caused by either low-light or over-/under-exposure conditions. Corresponding enhancement functions implemented using pretrained IAT models are applied before the images are forwarded to the continual learning classifier for final prediction.

and proposes a novel two-branch transformer architecture consisting of a pixel-wise local branch f and a global ISP-guided branch g . In the local branch f , the input image is mapped into a latent feature space using a series of pixel-enhancement modules (PEMs). Although the PEM structure follows the layout of a standard transformer block, the self-attention operator is replaced by depthwise convolution layers to maintain a lightweight design and reduce computational overhead. In the global branch g , the model employs transformer attention queries to estimate and regulate global ISP-related parameters, including the color transformation matrix and gamma correction coefficients. The learned queries adapt dynamically to diverse illumination conditions, enabling the model to simultaneously correct both under-exposed and over-exposed regions within a unified framework.

3. Methodology

The overall workflow of our proposed approach is shown in Figure 1. The key idea is to evaluate the robustness of continual learning under illumination uncertainty by introducing controlled low-light and exposure perturbations at inference time, and to assess whether an

automatic illumination calibration module (IAT) can restore performance prior to classification. The complete pipeline comprises four major stages: (i) continual learning training using clean data streams, (ii) simulation of uncertain illumination contexts, (iii) IAT-based image enhancement, and (iv) inference under uncertainty.

3.1. Continual Learning Training via the Avalanche Framework

We adopt the Avalanche framework to implement standard continual learning strategies and manage the task sequence during training. The original training dataset is partitioned into a stream of experiences, each corresponding to a distinct incremental learning step. During training, the continual learning model sequentially processes each experience, updating the model parameters using the selected CL strategy, including Experience Replay (ER), Elastic Weight Consolidation (EWC), or Synaptic Intelligence (SI).

This training stage is conducted exclusively on the clean training data, without any illumination perturbations or enhancement processing. Upon completion of all experiences, a final CL model is obtained which serves as the base classifier for subsequent evaluations under uncertain imaging conditions.

3.2. Simulation of Illumination Uncertainty

To emulate real-world acquisition variability that commonly degrades medical image quality, we synthesize a set of validation datasets exhibiting controlled illumination perturbations. These uncertain contexts are generated using the `ColorJitter` module from PyTorch, which modifies image brightness and contrast to simulate both low-light and exposure-related distortions.

Low-Light Simulation Low-light scenarios are created by adjusting image brightness using multiplicative scaling factors in ranges where both bounds are below 1, darkening the image to differing extents. Three tiers of illumination degradation are defined:

- Upper Low Light: brightness scaled within $[0.7, 0.9]$,
- Low Light: brightness scaled within $[0.4, 0.6]$, and
- Lower Low Light: brightness scaled within $[0.1, 0.3]$.

Over- and Under-Exposure Simulation Exposure artifacts are synthesized by modifying image contrast using scaling ranges spanning below and above 1. This combination generates both overexposed and underexposed appearances. Three exposure levels are simulated:

- Upper Exposure: $[0.8, 1.2]$,
- Exposure: $[0.5, 1.5]$, and
- Lower Exposure: $[0.3, 1.7]$.

To maintain a balanced evaluation set while introducing stochastic variability, each augmentation is wrapped inside the `RandomApply` function with a probability of 0.5. As a result, only half of the validation samples undergo perturbation, while the remainder preserve their original appearance.

Table 1: Summary of the PathMNIST dataset

Name	Data modality	Task(# classes)	Resolution	Samples	Training/Validation
PathMNIST	Colon Pathology	Multi-class (9)	28x28	107,180	89,996 / 10,004

3.3. IAT-Based Illumination Calibration

To counteract the harmful effects of illumination degradation prior to classification, we integrate the Illumination Adaptive Transformer (IAT) as a preprocessing module. Two pretrained IAT networks are employed:

- a low-light enhancement model specialized for brightness restoration, and
- a contrast correction model designed to compensate for over- and under-exposure.

Given a perturbed input image, the pipeline determines whether the degradation corresponds primarily to low-light or contrast distortion based on the applied augmentation type. The respective IAT model is then invoked via a wrapper function that applies the enhancement in a forward-pass manner. Both enhancement networks remain frozen during all experiments; they are used strictly for inference-time preprocessing without any integration into the CL optimization loop.

The output from either IAT branch yields enhanced images with improved brightness consistency and contrast normalization. These outputs are merged into a unified enhanced validation dataset, ensuring input distributions that more closely match those encountered during uncontaminated training.

3.4. Classification and Evaluation Under Uncertainty

Finally, the enhanced validation images are fed into the trained continual learning classifier for inference. By comparing performance across original corrupted images and enhanced counterparts, we quantify the effects of illumination uncertainty on CL robustness and the extent to which IAT preprocessing restores classification accuracy. This protocol allows us to assess the compatibility of illumination calibration with different CL strategies and determine its effectiveness in stabilizing prediction performance under real-world visual variability.

4. Experiments

4.0.1. DATASET

We conducted our experiments using the PathMNIST dataset from the MedMNIST benchmark suite (2), a standardized collection of biomedical image datasets specifically designed for multi-class medical image classification and machine learning evaluation. MedMNIST consolidates diverse datasets derived from multiple medical imaging modalities into a unified format, providing an accessible and reliable platform for developing and benchmarking learning algorithms in healthcare applications.

PathMNIST is a curated histopathology dataset collected from colorectal cancer tissue specimens. It comprises 107,180 individual RGB images, each resized to a standardized

Table 2: Hyperparameter Settings

Hyperparameter	Value
Training mini-batch size	500
Number of epochs	4
Evaluation mini-batch size	100
Optimizer	AdamW
Learning rate	0.001

spatial resolution of 28×28 pixels. The images are annotated into nine clinically relevant tissue categories, representing common histological structures observed in colorectal pathology. This diversity of tissue types introduces substantial visual variability, making the dataset well-suited for evaluating both classification accuracy and robustness in medical imaging models.

Motivated by realistic deployment scenarios in which new tissue categories may become available over time, we consider the class-incremental continual learning setting throughout our experiments. Using the Avalanche framework, PathMNIST is restructured into a sequential data stream containing three consecutive learning experiences, with each experience introducing three previously unseen classes. This setup enables systematic evaluation of catastrophic forgetting and knowledge retention as the model must continuously expand its label space while maintaining performance on earlier categories.

4.1. Experimental Setup

In our experiments, we selected a unified set of training hyper-parameters to ensure stable optimization and enable fair comparisons across all evaluated continual learning strategies. The same base configuration was applied to each method to eliminate performance bias arising from inconsistent optimization settings. The general hyper-parameters are summarized in Table 2. In addition to the general training configuration, strategy-specific hyper-parameters were defined for each continual learning baseline:

- **Experience Replay (ER):** memory buffer size = 200.
- **Elastic Weight Consolidation (EWC):** regularization strength $\lambda_{\text{EWC}} = 0.1$.
- **Synaptic Intelligence (SI):** regularization strength $\lambda_{\text{SI}} = 0.1$.

Here, λ_{EWC} and λ_{SI} control the degree to which parameter updates are constrained to preserve knowledge from previously learned tasks.

All hyperparameter values were determined through preliminary validation experiments and a small-scale sensitivity analysis aimed at identifying settings that provide a reasonable balance between knowledge retention and forward adaptation. The Experience Replay buffer size was selected to maintain a compact yet representative subset of past samples while avoiding excessive memory overhead. Similarly, the regularization coefficients for EWC and SI were tuned to sufficiently restrict destructive parameter drift without overly suppressing learning capacity on new tasks.

Table 3: Top-1 Accuracy Performance Comparison across different CL strategies and conditions with IAT(Our method) and without IAT

Condition	Replay	EWC	SI	SI + IAT(Ours)
Original(Upper bound)	0.676	0.760	0.870	-
Upper Low Light	0.510	0.580	0.660	0.780
Low Light	0.460	0.520	0.590	0.680
Lower Low Light	0.400	0.460	0.520	0.580
Upper Exposure	0.600	0.680	0.770	0.840
Exposure	0.520	0.590	0.670	0.720
Lower Exposure	0.430	0.490	0.560	0.600

While the selected hyperparameters yielded consistently strong performance across all evaluated scenarios, we note that optimal configurations may vary depending on dataset scale, class complexity, and task ordering. Automated hyperparameter optimization or adaptive parameter scheduling constitutes a promising direction for future work to further enhance performance under diverse continual learning settings.

4.2. Results and Discussion

Table 3 summarizes the Top-1 Accuracy achieved by each continual learning (CL) strategy under both standard lighting conditions and a range of simulated illumination uncertainties, including the proposed integration of the Illumination Adaptive Transformer (IAT) with Synaptic Intelligence (SI). On the original validation set, which represents the upper bound of achievable performance, SI attains the highest accuracy of 87.0%, substantially outperforming both EWC (76.0%) and Replay (67.6%). This result highlights the strong capability of SI to preserve previously learned knowledge while assimilating new classes in the class-incremental setting.

Under all uncertain lighting conditions, classification accuracy deteriorates for every CL strategy. Performance degradation is notably more severe in low-light conditions than under exposure variations. For example, the accuracy of SI decreases from 87.0% on clean data to 52.0% under the most extreme low-light condition, whereas the reduction is less pronounced for severe exposure distortions, where performance drops to 56.0%. Across the full range of evaluated conditions, SI consistently outperforms EWC and Replay, and the relative performance gaps between methods remain largely stable. This consistency indicates that SI maintains its comparative advantage even as visual perturbations intensify.

Integrating the Illumination Adaptive Transformer with SI yields substantial accuracy gains across all uncertain contexts. In low-light scenarios, IAT improves SI performance from 66.0% to 78.0% in the Upper Low Light condition, corresponding to a 12 percentage-point increase. For exposure distortions, the integration enhances accuracy from 77.0% to 84.0% under the Upper Exposure condition, leading to a 7 percentage-point improvement. Importantly, the SI+IAT configuration achieves the highest accuracy across every tested uncertainty level, as emphasized by the bold values reported in Table 3.

A clear monotonic trend emerges in which classification accuracy decreases as the severity of illumination perturbations increases from “Upper” to “Lower” settings. Nevertheless, the proposed SI+IAT approach consistently mitigates this degradation, maintaining stronger performance relative to all baseline strategies throughout the spectrum of uncertainty levels. Although the enhancements provided by IAT are substantial, even the best results under extreme conditions remain below the clean-data upper bound of 87.0%, indicating that severe illumination distortions continue to pose a significant challenge. Future work may explore tighter integration between enhancement modules and continual learning objectives, or the development of robustness-aware CL algorithms, to further close this performance gap.

5. Conclusion

This study demonstrates the practical effectiveness of continual learning strategies for medical image classification under both standard and uncertain imaging conditions. The integration of the Illumination Adaptive Transformer proves effective in mitigating the adverse impact of data-quality uncertainty, enhancing robustness across a wide range of simulated lighting distortions. These findings indicate that illumination-aware preprocessing is well suited to continual learning scenarios, where models must operate over sequential tasks while facing potential variability in visual quality and acquisition conditions.

Despite these improvements, performance degradation under extreme uncertainty remains evident, highlighting the need for further advances to fully address the complexities of real-world imaging variability. Future work will expand this study to incorporate more diverse datasets featuring naturally occurring illumination variations and acquisition inconsistencies, enabling more comprehensive evaluation of robustness and paving the way toward clinically reliable continual learning systems.

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